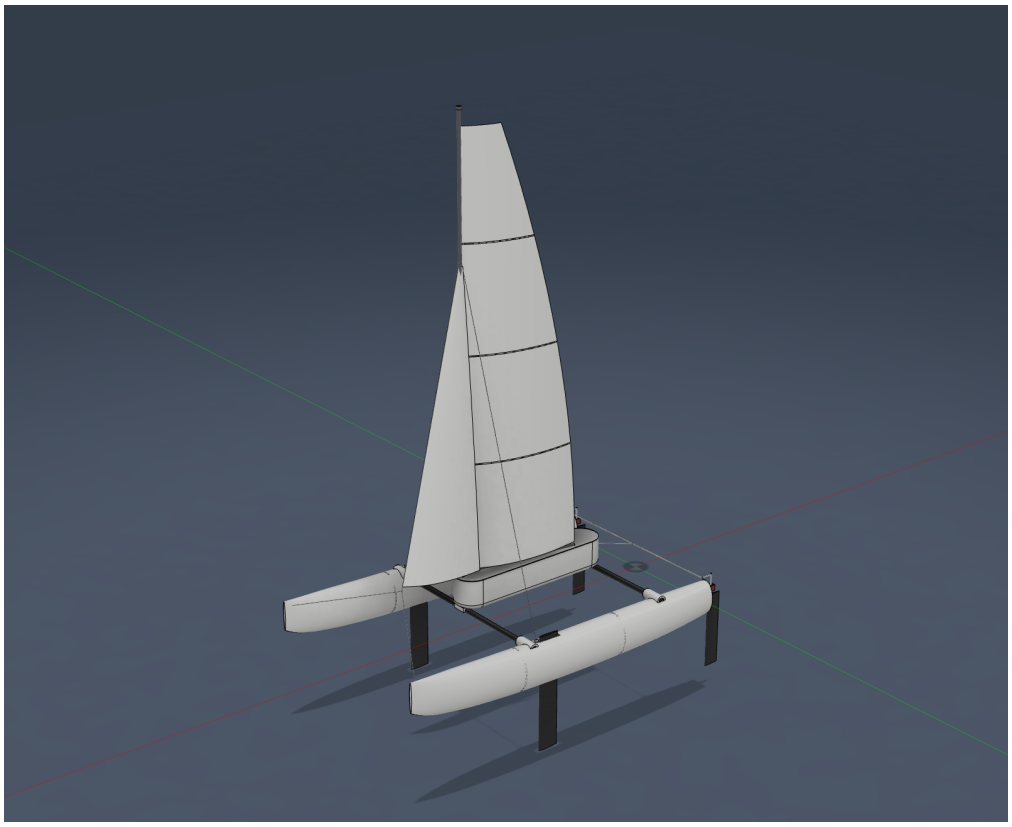
RC Mini Cat 600

Build Manual and Costed Parts List

A simplified 600 mm version of the RC Foiling Catamaran project



Length / overall beam	600 mm / 394 mm
Sailing mass (estimate)	755 g
Sail area (main + jib)	0.143 m^2
Radio	2 channels used: rudder + one sail winch
Parts cost (USD, Sep 2026)	$257.37

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1 What you are building

The Mini Cat is the smallest step toward the 1 m RC catamaran: the same hull lines (scaled 0.6 in length and 0.68 in section), the same beach-cat sail plan (square-top main with 3 full-length battens, jib on a forestay bridle), and the same printed-hull construction (0.8 mm PETG skin, every print segment a sealed compartment). It keeps what matters for learning to build and sail the big boat and drops what makes the big boat hard.

	1 m boat (build-first v1.2)	Mini Cat 600 (this manual)
Sail controls	3 servos: jib sheet, traveler, mainsheet tension	1 winch pulls both sheets on a stepped drum
Mast	rotating wing mast	fixed 6 mm carbon tube
Main battens	8	3
Mainsheet	traveler car on the rear beam	fixed fairlead on the box lid
Hull pieces to print	10 (5 per hull)	6 (3 per hull)
Radio channels used	4	2
Parts cost	$381.02	$257.37

Predicted numbers (from the project's analysis code; not yet measured on the water)

- Floats with the hulls 24 mm deep and 50 mm of freeboard at mid-length; level trim (centre of buoyancy 262 mm, centre of gravity 258 mm from the transom).
- Each hull alone could float 2.05 times the boat's weight (design target 2 or more).
- Speed on a beam reach: 0.95 m/s in 4 kn of wind, 2.13 m/s in 8 kn. The rig must be eased (depowered) above about 8 kn close-reaching and 11 kn on a beam reach.
- Steering: at most 1.6° of rudder is needed of the 19.9° the linkage gives, leaving at least 86 % margin before the rudders stall.

2 What it costs

Group	Cost
Radio	$72.98
Servos	$13.72
Power	$16.75
Printed	$13.99
Carbon	$43.36
Hardware	$20.52
Sails	$42.84
Rigging	$14.43
Adhesives	$18.78
Total parts	$257.37

Prices were read on the retailers' pages on 23 September 2026 and re-checked on 24 September 2026 (links in Appendix C). Not included: shipping (HobbyTown charges a $3.99 flat rate; other stores

vary), sales tax, a NiMH charger for the eneloop cells, the 3D printer, paint, and hand tools. The FS-i6X radio is the largest single item and is reused unchanged for the 1 m boat.
Options: EcoPower 640T waterproof micro servo ($29.99, instead of MG90S); FlySky FS-GT2E + FS-A3 (car radio) ($38.99, instead of FS-i6X; throttle trigger springs back to centre).

3 Parts list

Group	Item	Used for	Qty	Unit	Total
Radio	FlySky FS-i6X + FS-iA6B receiver	CH1 rudder, CH3 sails (non-centering stick); reused on the 1 m boat	1	$69.99	$69.99
Radio	AA alkaline x4	transmitter power	1	$2.99	$2.99
Servos	JX W4505-2T sail winch servo	both sheets on the stepped drum	1	$11.13	$11.13
Servos	MG90S metal-gear micro servo	rudders via pushrod + tie bar	1	$2.59	$2.59
Power	Panasonic eneloop AA 2000 mAh, 4-pack	receiver pack (4.8 V)	1	$12.75	$12.75
Power	Traxxas 4-cell AA holder with switch (TRA1523)	receiver battery box + on/off	1	$4.00	$4.00
Printed	PETG filament 1 kg	hulls, box, lid, rudders, boards, drum (470 g used)	1	$13.99	$13.99
Carbon	Carbon tube 10 x 8 mm x 1 m	two 360 mm crossbeams	1	$16.15	$16.15
Carbon	Carbon tube 6 x 4 mm x 1 m	mast (800 mm)	1	$15.25	$15.25
Carbon	Carbon tube 4 mm OD x 1 m	boom (210 mm)	1	$8.49	$8.49
Carbon	Carbon rod 1.2 mm x 1 m	3 battens	1	$3.47	$3.47
Hardware	Music wire 0.125 in	rudder stocks	1	$3.99	$3.99
Hardware	Music wire 0.093 in	tie bar, tiller arms	1	$3.99	$3.99
Hardware	Music wire 0.047 in	rudder pushrod, sheet hooks	1	$3.99	$3.99
Hardware	Du-Bro micro pushrod system	clevis + tube for the rudder pushrod	1	$8.55	$8.55
Sails	Icarex PC31 ripstop polyester, 1 m	main + jib (0.145 m2)	1	$21.99	$21.99
Sails	Ripstop sail tape 2 in, 10 ft	luff, batten pockets, corners	1	$20.85	$20.85
Rigging	Braided PE line 30 lb	stays, bridle, sheets	1	$14.43	$14.43
Adhesives	BSI 5-minute epoxy 4.5 oz	hull segment joints, beams, fittings	1	$13.99	$13.99
Adhesives	BSI gap-filling CA 1/2 oz	small parts, knots	1	$4.79	$4.79
	Total				$257.37

Tools and shop supplies (assumed on hand)

3D printer with a 220 mm bed that prints PETG; hobby knife and a steel rule; drill bits 1.2, 1.5, 2.5 and 3.2 mm (a pin vise is enough); side cutters and round-nose pliers for bending music wire; 240-grit sandpaper; masking tape; a fine-tip soldering iron or a hot needle to melt clean holes in the sailcloth; small screwdrivers for servo screws; a bucket or sink for leak tests; a kitchen scale.

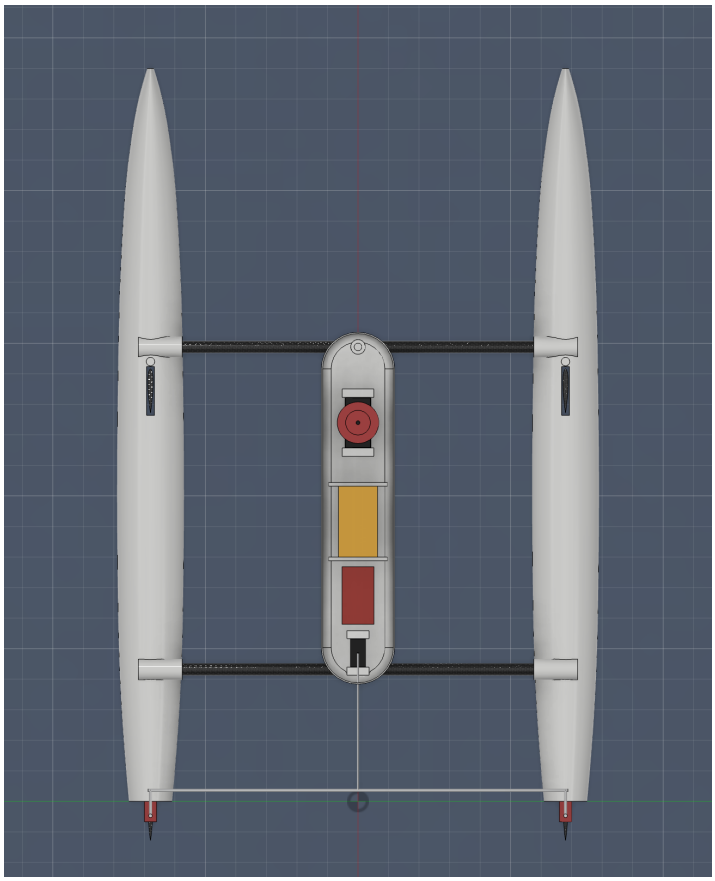
4 Printing

Settings used for the design mass: PETG, 0.4 mm nozzle, 0.2 mm layers, 2 perimeters (0.8 mm walls) for hulls, box and lid, 5 top and 5 bottom layers, 0 % infill for the hull segments (they are closed shells: each segment prints as a sealed compartment ending in half of a 2 mm bulkhead). Rudders, daggerboards and the drum are solid parts: 3 perimeters and 40 % infill. About 462 g of the 1 kg spool is used.

Part	Qty	g each	Size (mm)	Orientation / notes
Hull Port Seg 1 / 2 / 3	1 each	44 / 56 / 38	200 x 55 x 78	stand on the joint (bulkhead) face
Hull Stbd Seg 1 / 2 / 3	1 each	44 / 56 / 38	200 x 55 x 78	mirror of port
Box Half 1 / 2	1 each	37 / 42	144 x 60 x 56	open side up
Box Lid Half 1 / 2	1 each	10 / 10	144 x 60 x 1	flat
Daggerboard	2	27	200 x 36 x 4.3	flat; 3 perimeters, 40 % infill
Rudder Blade	2	10	130 x 27 x 3.2	flat; 3 perimeters, 40 % infill
Gudgeon	2	2	18 x 10 x 10	any
Winch Drum (30 / 16.2 mm)	1	7	34 x 34 x 10.4	large flange down

Files: `mini/stl/` (STL, one file per part), `mini/cad/RC_Mini_Cat_600.step` and `.f3d` (full assembly).

5 Build steps



Layout from above (lid and sails hidden): mast step, winch with stepped drum, battery, receiver, rudder servo, pushrod to the tie bar.

1. Hulls

1. Sand the joint faces of the three segments of each hull flat on sandpaper laid on a table.
2. Dry-fit: the segments must line up on the outside. Epoxy segment 1 to 2 and 2 to 3 face to face. Hold them against a straight edge (a steel rule taped to the table) while the epoxy cures.
3. Leak test: push each hull under water for 30 seconds. A stream of bubbles marks a leak: dry, then seal it with a drop of CA or epoxy. Because every segment is its own sealed compartment, a small leak floods only one segment.
4. Drill the chainplate eye: a 1.5 mm hole fore-aft through the small boss on the deck, 360 mm from the transom. Drill a 1.2 mm hole into each bow tip and epoxy in a small loop of 0.047 in music wire for the bridle.

Check: each finished hull weighs about 145 g (design value including glue) and floats level.

2. Crossbeams and frame

1. Cut the 10 mm carbon tube into two 360 mm beams (score around with a knife, then snap; wear a mask when sanding carbon).
2. Epoxy both beams into the sockets of one hull. Slide the other hull on, lay the frame on a flat table and measure the two diagonals (front-left beam end to rear-right, and the reverse). Shift until they are equal within 1 mm, then epoxy the second hull.

Check: hull centre lines are 340 mm apart at both beams.

3. Rudders and steering linkage

1. Drill a 3.2 mm hole down the pivot line of each rudder blade, 6.75 mm behind the leading edge (25 % of the 27 mm chord). This balanced pivot keeps the servo load tiny.
2. Push a 0.125 in music-wire stock through each blade and its gudgeon, epoxy the blade to the wire (not the gudgeon), and epoxy the gudgeon to the transom.
3. Bend two tiller arms from 0.093 in wire: 20.75 mm from the stock to the tie-bar hole. Keep the arms level, about 6 mm above the deck, so they clear it at full rudder. Join them with a straight 0.093 in tie bar at the transoms so the rudders stay parallel.
4. Run a 0.047 in pushrod from the rudder servo horn (hole 10 mm from the centre) through the slot in the rear wall of the box to the middle of the tie bar, using the Du-Bro clevis at the servo end.

Check: with the servo centred both rudders point straight; full servo travel gives about $\pm 20^\circ$ of rudder.

4. Electronics box

1. Epoxy the two box halves together, and separately the two lid halves.
2. Lower the sail winch into the well between the front rails (output shaft up) until its mounting ears rest on the rails, and set the MG90S between the rear rails the same way. Lead each servo wire out sideways through the notch at the foot of a rail. Drill 1.5 mm pilot holes through the ear holes and fix both servos with the screws supplied with them.

   Check: each case hangs from its ears, clear of the floor. The rails are drawn from the Feetech FS5115M drawing because JX does not publish the winch's ear height: if the winch case touches

the bottom of the well before its ears reach the rails, put a shim (a scrap of 1 mm plastic) under each ear.

3. Put four charged eneloop AA cells in the Traxxas holder and set it in the battery cradle; stick the receiver on the floor between the rudder servo and the battery with double-sided tape.
4. Plug in: rudder servo to CH1 (right stick, self-centring); sail winch to CH3 (left stick, which stays where you leave it); battery lead to a spare channel (for example CH6). Keep both receiver antennas straight and near the top of the box.
5. Glue the printed stepped drum to the winch's round horn and screw it on.
6. Snap the box onto the two beams (the saddles clip over the 10 mm tubes). Seal the lid seam with electrical tape before sailing.

5. Mast and standing rigging

1. Cut the 6 mm tube to 799 mm. Plug the top with epoxy. Mark the hounds (where the stays attach) 574 mm above the mast foot and whip a tight binding of line there, glued with CA.
2. Stand the mast in the socket on the box floor through the lid hole.
3. Tie the stays from braided line (lengths between knots): forestay 529 mm from the hounds to the bridle point; two bridle legs of 206 mm from the bridle point to the bow loops; two shrouds of 602 mm from the hounds to the chainplate eyes.
4. Cut the 4 mm tube to a 207 mm boom. Tie a loose loop of line through its front end around the mast so the boom sits about 5 mm above the lid and swings freely.

Check: the mast stands straight from the side and from behind; all stays are just tight.

6. Sails

1. Print `mini/manual/sail_patterns_1to1.pdf` at 100 % (large-format print, or tile it). The 100 mm scale bar must measure 100 mm.
2. Lay the Icarex on the pattern and cut the main and jib with a hot knife or scissors. The main needs the extra 25 mm on the luff for the mast sleeve.
3. Main: fold the 25 mm sleeve and close it with sail tape so it slides over the mast. Cut three battens from 1.2 mm carbon rod (185, 155, 111 mm) and tape them on at 178, 357, 536 mm above the foot.
4. Jib: fold a 10 mm sleeve along the 508 mm luff around the forestay line.
5. Reinforce the head, tack and clew of both sails with two layers of sail tape and melt a 2 mm hole in each clew.

7. Sheets and drum

Both sheets end on the one drum, so one stick moves both sails.

- Mainsheet: from the boom end, down through the lid fairlead 165 mm from the transom, to the 30 mm groove.
- Jib sheet: from the jib clew, down through the fairlead 355 mm from the transom, to the 16.2 mm groove.

- Wind both sheets the same way. From sheeted in to fully eased the mainsheet pays out 153 mm and the jib sheet 83 mm. Both take 1.63 drum turns, which is why the grooves have different diameters.

8. Radio setup

1. Bind the receiver to the transmitter (follow the FlySky manual) and select an airplane model.
2. Set the CH3 end points so full stick travel turns the drum 1.63 turns. If the winch makes 2 turns at 100 %, that is about 82 %. Reverse CH3 if needed so that stick down means sheets in.
3. Set CH1 so the right stick steers the boat the way you expect: stick right, bow turns right.
4. Use the radio's failsafe function: on signal loss, rudders centred and sheets eased. An eased cat stops and drifts instead of sailing away.

9. Balance and float test

1. Rig the boat. Balance it on two fingers under the beams: it should balance about 258 mm forward of the transoms. If not, move the battery.
2. Float it in a tub: the hulls should sit level, about 24 mm deep. Leave it 10 minutes and check for water in the box.

10. First sail

- Pick 3 to 8 kn of wind, a small pond, and a way to retrieve the boat that does not involve swimming (a friend's kayak, or a long pole and line). A capsized catamaran does not right itself.
- Start on a beam reach with the sheets half out. Sheet in until the boat accelerates; ease when a hull starts to lift.
- Above about 8 kn, ease the sheets to keep the windward hull down.
- Watch the bows. A catamaran more often trips over its bows (pitchpoles) than capsizes sideways. The analysis puts the bow-burying limit at about 7.5 / 8.5 / 10 / 13.5 kn of wind at 60 / 90 / 120 / 150° off the wind. Near those winds, ease the sheets as soon as the leeward bow starts to dig.
- If the boat keeps turning away from the wind, sheet the jib less (the jib rides 4° wider than the main on the drum) or move the mast step aft. If it keeps turning into the wind, do the opposite.

6 Troubleshooting

Symptom	Fix
Water inside a hull segment	Leak test that segment again; seal with CA; paint the outside with thinned epoxy
Winch stalls or jitters	Check the battery charge; reduce the CH3 end points; make sure no line is caught under the drum
Rudders do not centre	Re-bend the pushrod or adjust the clevis; set CH1 sub-trim
Boat will not tack	Sail faster before turning; turn gradually with full rudder; ease the jib during the turn
Nose dives in gusts	Ease both sheets earlier; move the battery aft

A Engineering numbers

Hydrostatics (fresh water): mass 755 g; waterline at z=-6.8 mm on the design reference; hull volume 1546 cm^3 per hull; maximum righting moment 1.20 N m from the heeled-hull calculation (largest righting arm 162 mm at 8°, capsize near 80°); largest bow-down moment the hulls can resist 0.82 N m on two hulls, 0.58 N m on one; heeling arm 0.42 m.
Predicted boat speed in m/s (* = sails eased to keep the windward hull down):

True wind (kn)	TWA 60°	TWA 90°	TWA 105°
2	0.39	0.43	0.41
4	0.89	0.95	0.89
6	1.39	1.51	1.41
8	1.87*	2.13*	2.07
10	2.01*	2.13*	2.13*
12	2.11*	2.13*	2.13*
14	2.11*	2.13*	2.13*

These come from the same velocity-prediction code as the 1 m boat. Hull friction uses the ITTC-57 line; wave drag uses the Michell thin-ship integral with catamaran interference; sail forces use a lift/drag model. The code and its validation checks are in `analysis/vpp.py` and `mini/mini_analysis.py`. They are predictions until measured with GPS on the water.

B Parts cost of the 1 m build-first boat

Group	Item	Used for	Qty	Unit	Total
Radio	FlySky FS-i6X + FS-iA6B receiver	4 channels used: rudder, traveler, jib, mainsheet	1	$69.99	$69.99
Radio	AA alkaline x4	transmitter power	1	$2.99	$2.99
Servos	JX W4505-4T sail winch servo	traveler (about 231 mm of car travel, rig model)	1	$11.13	$11.13
Servos	JX W4505-2T sail winch servo	jib sheet (about 93 mm of sheet, rig model)	1	$11.13	$11.13
Servos	Feetech FS5115M standard servo	mainsheet tension	1	$32.06	$32.06
Servos	MG90S metal-gear micro servo	rudders via pushrod + tie bar	1	$2.59	$2.59
Power	Panasonic eneloop AA 2000 mAh, 4-pack	receiver pack (4.8 V)	1	$12.75	$12.75
Power	Traxxas 4-cell AA holder with switch (TRA1523)	receiver battery box + on/off	1	$4.00	$4.00
Printed	PETG filament 1 kg	hulls, pod, lid, rudder-servo bracket and cover, rudders, boards, gudgeons, tracks, cars, mast fairing and fittings (1.5 kg used, 1 m manual print list)	2	$13.99	$27.98
Carbon	Carbon tube 16 x 14 mm x 1 m	two 600 mm crossbeams	2	$16.28	$32.56
Carbon	Carbon tube 12 x 10 mm x 1 m	two-piece mast (1.3 m)	2	$12.26	$24.52

Group	Item	Used for	Qty	Unit	Total
Carbon	Carbon tube 10 x 8 mm x 1 m	mast joiner spigot (150 mm) and pod beam (600 mm)	1	$16.15	$16.15
Carbon	Carbon tube 8 x 7 mm x 1 m	boom (345 mm)	1	$17.20	$17.20
Carbon	Carbon rod 1.2 mm x 2 m	8 battens	1	$5.25	$5.25
Hardware	Music wire 0.156 in	rudder stocks, tie bar	2	$4.99	$9.98
Hardware	Music wire 0.093 in	rudder pushrod, track end-stop turning pins, gooseneck pin, boom sheet eye	1	$3.99	$3.99
Hardware	Du-Bro micro pushrod system	clevises	1	$8.55	$8.55
Sails	Icarex PC31 ripstop polyester, 1 m	main + jib (0.40 m2)	1	$21.99	$21.99
Sails	Ripstop sail tape 2 in, half roll	luff, batten pockets, corners	1	$24.00	$24.00
Rigging	Braided PE line 30 lb	stays, bridle, sheets	1	$14.43	$14.43
Adhesives	BSI 5-minute epoxy 9 oz	hull joints, beams, fittings	1	$22.99	$22.99
Adhesives	BSI gap-filling CA 1/2 oz	small parts	1	$4.79	$4.79
	Total				$381.02

If you build the mini first, the radio and the NiMH pack carry over, which lowers the 1 m boat's extra cost by $89.73.

C Parts cost of the v2.3 foiling kit

The kit is sized for the v2.3 foils in the CAD model: NACA 4412 wings 180 x 50 mm at +1.5°, NACA 0012 elevators 160 x 28 mm at −0.5° (nose down), and 60 mm-chord struts. The foiling rudders end 18 mm lower than the build-first ones, in a printed trim pod: bond a 8 mm extension (`stl/Foiling_Rudder_Extension_x2.stl`) and then the pod (`stl/Elevator_Trim_Pod_x2.stl`, 45 x 9 x 14 mm) under each blade. The elevator passes through the pod's slot, pivots on a 2 mm carbon pin at its quarter chord, and is clamped by two M2 set screws 12 mm behind the pin (one from above, one from below): one turn moves it about 1.9°, so it can be trimmed on the water. Printable cores with 0.5 mm recesses for the carbon caps are in `stl/` (`Foil_Core_*`, `Foil_Flap_x2`). The foils are printed PETG cores with unidirectional carbon caps (4 layers per face), laminated with epoxy; the caps take about 5.0 m of the 7.3 m of tape listed. The kit also includes flap hinges, the ride-height wands and their linkage. It is added to the 1 m boat after the build-first version sails.

Group	Item	Used for	Qty	Unit	Total
Printed	PETG filament 1 kg	L-foil, elevator and flap cores, trim pods, ride-height linkage (290 g, 1 m manual print list)	1	$13.99	$13.99
Carbon	Unidirectional carbon tape 2 in x 12 ft	spar caps: 4 layers per face on struts, wings, elevators	2	$19.95	$39.90
Adhesives	EL2 laminating epoxy 500 g	wet-out of the caps	1	$20.70	$20.70
Carbon	Carbon rod 1.2 mm x 1 m	flap hinge pins	1	$3.47	$3.47
Carbon	Carbon rod 2 mm x 1 m	two 300 mm ride-height wands, elevator pivot pins, wand and bellcrank axles	1	$8.85	$8.85
Hardware	Music wire 0.047 in	wand-to-flap pushrods	1	$3.99	$3.99
Hardware	Du-Bro micro pushrod system	clevises for the flap linkage	1	$8.55	$8.55

Group	Item	Used for	Qty	Unit	Total
Hardware	M2 x 6 mm stainless set screw, cup point (Bolt Depot 8697)	elevator trim screws, two per trim pod	4	$0.43	$1.72
	Total				$101.17

D Price sources (read 23-24 September 2026)

- HobbyTown: https://www.hobbytown.com/flysky-i6x-6channel-afhds-2a-air-radio-system-w-ia6b-receiver-fsy-i6xia6b/p1674181
- HobbyTown (EcoPower AA x4 add-on): https://www.hobbytown.com/flysky-fsgt2e-2channel-transmitter-w-fsa3-receiver-fsy-fs-gt2ea3/p1592822
- AliExpress: listed $5.79 at 48% off; full price used: https://www.aliexpress.com/item/1005003265131874.html
- Waveshare: https://www.waveshare.com/mg90s-servo.htm
- MicroBattery: https://www.microbattery.com/panasonic-bk-3mcca4ba-eneloop-aa-2100-cycle-ni-mh-pre-charged-rechargeable-batteries-4-pack80.html
- HobbyTown: https://www.hobbytown.com/traxxas-4cell-aa-battery-holder-w-switch-cover-tra1523/p49945
- Elegoo US: https://us.elegoo.com/products/rapid-petg-filament-1-75mm-colored-1kg
- CST CS-525 1M: https://www.cstsales.com/Carbon_Fiber_Tubes.html
- CST CS-316 1M: https://www.cstsales.com/carbon_rods.html
- K&S 0.125 in x 12 in: https://ksmetals.com/collections/12-long-music-wire
- HobbyTown: https://www.hobbytown.com/dubro-pushrod-system-micro-dub847/p29360
- Jones Airfoils, per metre (138 cm wide): https://jonesairfoils.com/products/icarex
- kiteboarding.com: $15.00 base + $5.85 for 10 ft: https://kiteboarding.com/proddetail.asp?prod=ripstop_2inch
- Walmart, PowerPro 30 lb x 150 yd (read 2026-09-24; the original eBay listing had ended): https://www.walmart.com/ip/PowerPro-Braided-Fishing-Line-30lb-x-150yd-ylw-21100300150Y/17117880
- HobbyTown: https://www.hobbytown.com/bob-smith-industries-quickcure-5-minute-epoxy-4-1-2oz-bsi201/p37081
- HobbyTown: https://www.hobbytown.com/bob-smith-industries-instacure-gap-filling-ca-1-2oz-bsi106/p37055
- HobbyTown: https://www.hobbytown.com/ecopower-640t-13g-waterproof-metal-gear-digital-sub-micro-servo-ecp-640t/p859667
- Pololu: https://www.pololu.com/product/3426
- Wind Catcher RC (price from search listing, page not reachable from here): https://windcatcherrc.com/product/carbon-fiber-tube-16mm-x-14mm-x-1000mm/
- RMRC: https://www.readymaderc.com/products/details/carbon-fiber-tube-12mm-x-10mm-1m-long
- HobbyTown: https://www.hobbytown.com/bob-smith-industries-quickcure-5-minute-epoxy-9o-bsi202/p37082
- CST CU-379: 2 in unidirectional carbon, 0.005 in, 12 ft: https://www.cstsales.com/heat_tack.html
- Easy Composites EL2, 500 g kit: https://www.easycomposites.us/el2-epoxy-laminating-resin

- Bolt Depot, per piece (read 2026-09-25): `https://boltdepot.com/Product-Details?product=8697`

Two prices carry extra uncertainty: the JX winch (AliExpress listed it on sale; the full price is used here) and the 16 x 14 mm carbon tube (price taken from a search listing because the store's page was not reachable).